CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2022 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Obtain a real root of the equation f(x) = x³ - x - 1 = 0 using Bisection method correct to three decimal places.**

Solution: $f(x) = x^3 - x - 1$.

$f(1) = 1 - 1 - 1 = -1 < 0$, $f(2) = 8 - 2 - 1 = +5 > 0$. Root lies in $[1, 2]$.

Iter 1: $x_1 = 1.5 \Rightarrow f(1.5) = 3.375 - 1.5 - 1 = 0.875 > 0 \Rightarrow [1, 1.5]$.

Iter 2: $x_2 = 1.25 \Rightarrow f(1.25) = 1.953 - 1.25 - 1 = -0.297 < 0 \Rightarrow [1.25, 1.5]$.

Iter 3: $x_3 = 1.375 \Rightarrow f(1.375) = 2.599 - 1.375 - 1 = +0.224 > 0 \Rightarrow [1.25, 1.375]$.

Iter 4: $x_4 = 1.3125 \Rightarrow f(1.3125) = -0.0515 < 0 \Rightarrow [1.3125, 1.375]$.

Iter 5: $x_5 = 1.3438 \Rightarrow f(1.3438) = +0.0826 > 0 \Rightarrow [1.3125, 1.3438]$.

Iter 6: $x_6 = 1.3281 \Rightarrow f(1.3281) = +0.0145 > 0$.

Iter 7: $x_7 = 1.3203 \Rightarrow x_8 \approx 1.3247$.

Root: $\mathbf{x \approx 1.325}$.

Q2 (3M) **Define forward difference. Make forward difference table for the ordered pairs: (x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4).**

Definition: For equally spaced points $x_i = x_0 + i\,h$, the forward difference operator Δ is defined as:

> $\Delta y_i = y_{i+1} - y_i$

Higher-order differences: $\Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i$.

Forward Difference Table:

x	y	Δy	Δ²y	Δ³y	Δ&sup4;y
x_0	y_0	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$	$\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$	$\Delta^4 y_0 = \Delta^3 y_1 - \Delta^3 y_0$
x_1	y_1	$\Delta y_1 = y_2 - y_1$	$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$	$\Delta^3 y_1 = \Delta^2 y_2 - \Delta^2 y_1$	-
x_2	y_2	$\Delta y_2 = y_3 - y_2$	$\Delta^2 y_2 = \Delta y_3 - \Delta y_2$	-	-
x_3	y_3	$\Delta y_3 = y_4 - y_3$	-	-	-
x_4	y_4	-	-	-	-

Q3 (3M) **Find the value of dy/dx at x = 2.2 for: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).**

Solution: Here $x = 2.2$ is at the end of the table ($x_n = 2.2$, $h = 0.2$).

Backward differences at $x_n = 2.2$:

$\nabla y_n = 9.0250 - 7.3891 = 1.6359$

$\nabla^2 y_n = 1.6359 - (7.3891 - 6.0496) = 1.6359 - 1.3395 = 0.2964$

$\nabla^3 y_n = 0.2964 - (1.3395 - 1.0966) = 0.2964 - 0.2429 = 0.0535$

Newton's backward derivative formula at $x = x_n$:

$$f'(x_n) = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n\right]$$

$= \frac{1}{0.2}\left[1.6359 + \frac{0.2964}{2} + \frac{0.0535}{3}\right] = 5 \times [1.6359 + 0.1482 + 0.0178] = 5 \times 1.8019 = \mathbf{9.0095} \approx \mathbf{9.01}$.

Exact value ($e^{2.2}$): 9.0250.

Q4 (3M) **Solve using Gauss elimination: 3x + 2y + 3z = 18, 2x + y + z = 10, x + 4y + 9z = 16.**

Solution: Reordering rows with row 3 as first:

$$\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 2 & 1 & 1 & | & 10 \\ 3 & 2 & 3 & | & 18 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 0 & -7 & -17 & | & -22 \\ 0 & -10 & -24 & | & -30 \end{bmatrix}$

$R_3 \to R_3 - \frac{10}{7} R_2$: $-24 - \frac{10}{7}(-17) = \frac{-168 + 170}{7} = \frac{2}{7}$, and $-30 - \frac{10}{7}(-22) = \frac{-210 + 220}{7} = \frac{10}{7}$.

$\frac{2}{7} z = \frac{10}{7} \implies \mathbf{z = 5}$.

$-7y - 17(5) = -22 \implies -7y - 85 = -22 \implies -7y = 63 \implies \mathbf{y = -9}$.

$x = 16 - 4(-9) - 9(5) = 16 + 36 - 45 = \mathbf{7}$.

Answer: $\mathbf{x = 7,\ y = -9,\ z = 5}$.

Q5 (3M) **Solve $dy/dx = x + y^2$, $y(0) = 0$ using Picard's method.**

Solution: $y_0 = 0$, $x_0 = 0$.

$y^{(1)}(x) = 0 + \int_0^x (t + 0)\, dt = \mathbf{\frac{x^2}{2}}$.

$y^{(2)}(x) = \int_0^x \left(t + \frac{t^4}{4}\right) dt = \mathbf{\frac{x^2}{2} + \frac{x^5}{20}}$.

$y^{(3)}(x) = \int_0^x \left[t + \left(\frac{t^2}{2} + \frac{t^5}{20}\right)^2\right] dt = \mathbf{\frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160}}$.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Find a real root of $f(x) = x^3 - 5x + 3 = 0$ using method of false position.**

Solution: $f(1) = 1 - 5 + 3 = -1 < 0$, $f(2) = 8 - 10 + 3 = +1 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = \frac{1(1) - 2(-1)}{1 - (-1)} = \frac{1 + 2}{2} = 1.5$.

$f(1.5) = 3.375 - 7.5 + 3 = -1.125 < 0 \Rightarrow$ root $\in [1.5, 2]$.

Iter 2: $x_2 = \frac{1.5(1) - 2(-1.125)}{1 - (-1.125)} = \frac{1.5 + 2.25}{2.125} = \frac{3.75}{2.125} = 1.7647$.

$f(1.7647) = 5.4955 - 8.8235 + 3 = -0.328 < 0 \Rightarrow$ root $\in [1.7647, 2]$.

Iter 3: $x_3 = \frac{1.7647(1) - 2(-0.328)}{1 + 0.328} = \frac{2.4207}{1.328} = 1.8228$.

$f(1.8228) = -0.062 < 0$.

Iter 4: $x_4 = \mathbf{1.831}$.

Answer: $\mathbf{x \approx 1.831}$.

Q7 (7.5M) **Find the missing term in the following table: x = [1, 2, 3, 4], y = [1, 3, 9, y_3].**

Solution: Since 3 values are given, approximate by a 2nd degree polynomial, so $\Delta^3 y_0 = 0$:

$$(E - 1)^3 y_0 = y_3 - 3y_2 + 3y_1 - y_0 = 0$$

Substitute known values: $y_3 - 3(9) + 3(3) - 1 = 0 \implies y_3 - 27 + 9 - 1 = 0 \implies y_3 - 19 = 0$.

$$\mathbf{y_3 = 19}$$

Note: If sequence was 3^{x-1}, value would be 27, but by polynomial interpolation $\mathbf{y_3 = 19}$.

Q8 (7.5M) **Evaluate $\int_0^{(\pi/2)} \sqrt{(\sin x)}\, dx$ using Simpson's 1/3 rule with h = π/12.**

Solution: Interval $[0, \pi/2]$ divided into 6 equal parts with $h = \pi/12 = 15^\circ$.

x	0°	15°	30°	45°	60°	75°	90°
√(sin x)	0.0000	0.5087	0.7071	0.8409	0.9306	0.9828	1.0000

Simpson's 1/3 Rule:

$$I = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$y_0 + y_6 = 0 + 1 = 1.0000$
$4(0.5087 + 0.8409 + 0.9828) = 4(2.3324) = 9.3296$
$2(0.7071 + 0.9306) = 2(1.6377) = 3.2754$
Sum $= 1.0000 + 9.3296 + 3.2754 = 13.6050$.

$$I = \frac{\pi / 12}{3} \times 13.6050 = \frac{\pi}{36} \times 13.6050 \approx 0.087266 \times 13.6050 = \mathbf{1.1873}$$

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Use Newton-Raphson method to find a real root of $4e^{-x}\sin(x) - 1 = 0$ correct to 3 decimal places, lying between 0 and 0.5.**

Solution: $f(x) = 4e^{-x}\sin x - 1$.
$f'(x) = 4e^{-x}(\cos x - \sin x)$.
$f(0) = 4(1)(0) - 1 = -1 < 0$.
$f(0.5) = 4e^{-0.5}\sin(0.5) - 1 = 4(0.6065)(0.4794) - 1 = 1.1630 - 1 = +0.1630 > 0$.
Take $x_0 = 0.3$:
$f(0.3) = 4e^{-0.3}\sin(0.3) - 1 = 4(0.7408)(0.2955) - 1 = 0.8757 - 1 = -0.1243$.
$f'(0.3) = 4(0.7408)(\cos 0.3 - \sin 0.3) = 2.9633(0.9553 - 0.2955) = 2.9633(0.6598) = 1.9552$.
$x_1 = 0.3 - \frac{-0.1243}{1.9552} = 0.3 + 0.0636 = 0.3636$.

Iter 2: $f(0.3636) = 4e^{-0.3636}\sin(0.3636) - 1 = 4(0.6952)(0.3556) - 1 = 0.9888 - 1 = -0.0112$.
$f'(0.3636) = 4(0.6952)(0.9346 - 0.3556) = 2.7808(0.5790) = 1.6101$.
$x_2 = 0.3636 - \frac{-0.0112}{1.6101} = 0.3636 + 0.0070 = 0.3706$.

Iter 3: $f(0.3706) = 4e^{-0.3706}\sin(0.3706) - 1 \approx 0.0000 \implies \mathbf{x = 0.371}$.
Final Root: $\mathbf{x \approx 0.371}$.

Q10 (15M) **Using Lagrange's interpolation formula, find the form of the function y(x) from: (0, -12), (1, 0), (3, 12), (4, 24).**

Solution: $(x_0, y_0) = (0, -12), (x_1, y_1) = (1, 0), (x_2, y_2) = (3, 12), (x_3, y_3) = (4, 24)$.

Since $y_1 = 0$, the second term vanishes.

$L_0(x) = \frac{(x-1)(x-3)(x-4)}{(0-1)(0-3)(0-4)} = \frac{(x-1)(x-3)(x-4)}{-12}$.

$L_2(x) = \frac{x(x-1)(x-4)}{(3-0)(3-1)(3-4)} = \frac{x(x-1)(x-4)}{-6}$.

$L_3(x) = \frac{x(x-1)(x-3)}{(4-0)(4-1)(4-3)} = \frac{x(x-1)(x-3)}{12}$.

Therefore: $y(x) = L_0(x)(-12) + L_2(x)(12) + L_3(x)(24)$

$= (x-1)(x-3)(x-4) - 2x(x-1)(x-4) + 2x(x-1)(x-3)$

$= (x-1)[\,(x-3)(x-4) - 2x(x-4) + 2x(x-3)\,]$

$= (x-1)[\,x^2 - 7x + 12 - 2x^2 + 8x + 2x^2 - 6x\,] = (x-1)[\,x^2 - 5x + 12\,]$

$$\mathbf{y(x) = x^3 - 6x^2 + 17x - 12}$$

Verification: $y(0) = -12$, $y(1) = 0$, $y(3) = 27 - 54 + 51 - 12 = 12$, $y(4) = 64 - 96 + 68 - 12 = 24$. (Exact!)

Q11 (15M) **Use Simpson's 3/8 rule to evaluate $\int_0^1 1/(1+x^2)\,dx$ with n = 6 intervals. Hence find the approximate value of π.**

Solution: $h = 1/6$. Values: $y_0 = 1$, $y_1 = \frac{36}{37} \approx 0.9730$, $y_2 = \frac{36}{40} = 0.9000$, $y_3 = \frac{36}{45} = 0.8000$, $y_4 = \frac{36}{52} \approx 0.6923$, $y_5 = \frac{36}{61} \approx 0.5902$, $y_6 = \frac{36}{72} = 0.5000$.

Simpson's 3/8 Rule:

$$I = \frac{3h}{8}\,[\,(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3\,]$$

$y_0 + y_6 = 1.5000$

$3(0.9730 + 0.9000 + 0.6923 + 0.5902) = 3(3.1555) = 9.4665$

$2(0.8000) = 1.6000$

$I = \frac{3(1/6)}{8}\,[\,1.5000 + 9.4665 + 1.6000\,] = \frac{1}{16}\,[\,12.5665\,] = \mathbf{0.785406}$.

Since $\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4} \implies \pi \approx 4 \times 0.785406 = \mathbf{3.1416}$.

Q12 (15M) **Solve the system by Gauss-Seidel iterative method: 20x + y - 2z = 17, 2x + 20y - 2z = -18, 2x - 3y + 20z = 25.**

Solution: The system is strictly diagonally dominant:

$$x = \frac{17 - y + 2z}{20},\quad y = \frac{-18 - 2x + 2z}{20},\quad z = \frac{25 - 2x + 3y}{20}$$

Start with $(x_0, y_0, z_0) = (0, 0, 0)$.

Iter 1: $x_1 = \frac{17}{20} = 0.8500$

$y_1 = \frac{-18 - 2(0.85)}{20} = \frac{-19.7}{20} = -0.9850$

$z_1 = \frac{25 - 2(0.85) + 3(-0.985)}{20} = \frac{25 - 1.7 - 2.955}{20} = 1.0173$

Iter 2: $x_2 = \frac{17 - (-0.9850) + 2(1.0173)}{20} = \frac{20.0196}{20} = 1.0010$

$y_2 = \frac{-18 - 2(1.0010) + 2(1.0173)}{20} = \frac{-17.9674}{20} = -0.9984$

$z_2 = \frac{25 - 2(1.0010) + 3(-0.9984)}{20} = \frac{20.0028}{20} = 1.0001$

Iter 3: $x_3 = \frac{17 - (-0.9984) + 2(1.0001)}{20} = 1.0000$

$y_3 = \frac{-18 - 2(1) + 2(1)}{20} = -1.0000$

$z_3 = \frac{25 - 2(1) + 3(-1)}{20} = 1.0000$

Final Answer: $\mathbf{x = 1.000,\ y = -1.000,\ z = 1.000}$.

Q13 (15M) **Using fourth order Runge-Kutta method, find y(0.2) and y(0.4) given dy/dx = x + y with y(0) = 1 (step size h = 0.2).**

Solution: $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, $h = 0.2$.

Step 1: Compute y(0.2)

$$k_1 = 0.2(0 + 1) = 0.2000$$

$$k_2 = 0.2(0.1 + 1.1) = 0.2400$$

$$k_3 = 0.2(0.1 + 1.12) = 0.2440$$

$$k_4 = 0.2(0.2 + 1.244) = 0.2888$$

$$\Delta y = \frac{0.2000 + 2(0.2400) + 2(0.2440) + 0.2888}{6} = \frac{1.4568}{6} = 0.2428$$

> $$y(0.2) = 1 + 0.2428 = \mathbf{1.2428}$$

Step 2: Compute y(0.4) from $x_1 = 0.2$, $y_1 = 1.2428$

$$k_1 = 0.2(0.2 + 1.2428) = 0.2886$$

$$k_2 = 0.2(0.3 + 1.2428 + 0.1443) = 0.3374$$

$$k_3 = 0.2(0.3 + 1.2428 + 0.1687) = 0.3423$$

$$k_4 = 0.2(0.4 + 1.2428 + 0.3423) = 0.3970$$

$$\Delta y_1 = \frac{0.2886 + 2(0.3374) + 2(0.3423) + 0.3970}{6} = \frac{2.0450}{6} = 0.3408$$

> $$y(0.4) = 1.2428 + 0.3408 = \mathbf{1.5836}$$

Answer: $y(0.2) = \mathbf{1.2428}$, $y(0.4) = \mathbf{1.5836}$.
